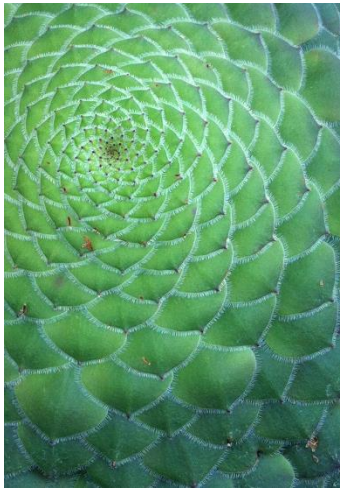
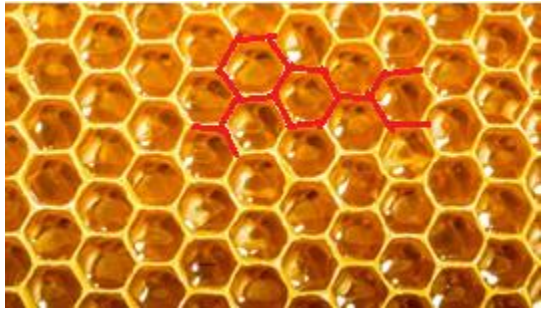
- **Where do we look for innovative ideas and resources?**



- Mother ‘nature’ is the greatest source of innovative ideas that human being can search for. It is an amazing experience when you observe the natural phenomenon. ‘Biomimicry’ is the field of study that focuses on nature and its behaviour. Human being have learned to mimic nature’s behaviour apply it in design.

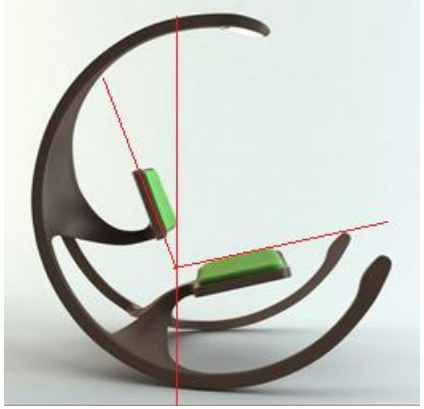
- Following are the basic observations-
- **Basic Form**
- **Functional Relationship** (how two parts relate to each other)
- **Behaviour** (how each part behaves in respect to other part)



Beehive Structure

Modular Structure

Functional Relationship and Behaviour



Does the seating device provides better comfort?

Does the above pen (A) design looks different from other regular pen design? Why? Any advantage?

- **Form**
- **Functional Relationship &**
- **Behaviour**

Always critically try to examine the above three…

Why the design is different from others?
Why we don't find such mug design in stores

(Product for Spastic People)

Does the above Bicycle design provides more comfort and creates more space?

In each case the functional relationship and their behaviour needs to be critically examined.

Ecology

Before we explore the magic of the Mother Nature it is important for us to sensitize ourselves with the richness of nature for ‘design thinking’.

- Abundance of Nature’s Paradigms has been known to human beings from the beginning of human civilizations. Mankind has observed nature with great deal of interest to learn its behaviour and mechanism. Since the dawn of man’s quest for knowledge and survival it became imperative to observe the great resources of nature. Human beings are surrounded by living and non-living beings. Naturally it is obvious that mankind would take help of such elements for their survival and comfort.

- **What is Biomimicry?**
- Lately '**Biomimicry'** (Design Paradigm) has become one of the most fascinating areas of studies to examining nature consciously, its **models, systems and processes.** In the process, man has learned to emulate or take inspiration from nature in order to solve their problems.
- Thus, nature has become one of the most inspiring teachers and helped us to invent new things. Man has learned to mimic the principles of nature in their favour to develop designs for 'safety and comfort'.

- Basically Biomimicry is asking-
- How would nature do this? The core idea is that nature is imaginative by necessity, has already solved many of the problems we are facing: energy, food production, climate control, non-toxic chemistry, transportation, self-defense system and a whole lot more.
- Nature has adopted various mechanism (internal) in order to survive.
- While human being creates new design (external- cold, hot, rain or desert) to survive instead of adopting the condition (which human being did in the very early stage). Because of the changing condition, adoption changes the organism.
- This led to a design-method where we try to translate these solutions in a systematic manner to producing innovative products.

Energy:
They are warm-blooded, and have a layer of fat, under the skin. This allows them to maintain a body temperature similar to human 36 °C (97 °F)—even in freezing conditions.

Reproduction:
There are five main modes of seedy dispersal: gravity, wind, ballistic (movement of their own), water, and by animals.

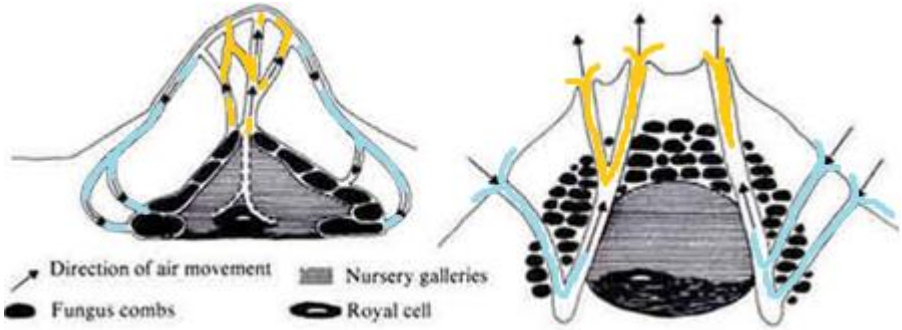
Climate control
Termite

Non-toxic chemistry-
Neem Tree

Atta- The Leaf Cutter Ant

Defense system- Self- protection

Transportation-
Ants are capable of carrying 50 times their own weight and up to 30 times their own volume. It is true. Among ants, the genus *Atta*, a leaf-cutter ant that lives in South America, holds the record. Surprisingly, these ants don’t eat the green leaves they collect from trees, which are cut with their own jaws— they use these leaves to feed the fungus colony they grow inside their nests. Ants feed on these fungi.



- **BIOMIMETIC ARCHITECTURE: Green Building in Zimbabwe Modeled After Termite Mounds**
- Inspired by Zimbabwean masonry and the self-cooling mounds of African termites, the Eastgate Centre in Harare, Zimbabwe, combines the best of green architecture and ecologically sensitive adaptation. Using biomimicry principles, the mid-rise building was designed by architect Mick Pearce in conjunction with engineers at Arup Associates. It has no conventional air-conditioning or heating, yet stays regulated year round with dramatically less energy consumption.

Land Snail

Collection of distinct useful characteristics-

Life of Butterfly

All butterflies and moths start out in the form of a caterpillar. Some, such as the monarch butterfly, form a cocoon to protect themselves as they become a pupa. When they are fully grown they come out of the pupa as an adult butterfly.

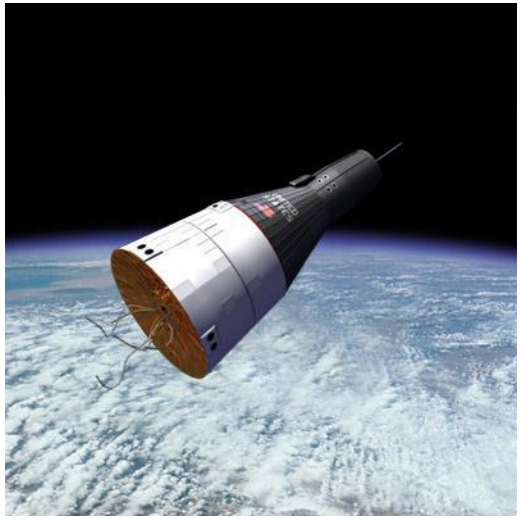
- **Interesting facts about Snails**

  Some snails hibernate. Some species of snails actually hibernate during the colder months of the year. They cover their bodies with a thin layer of mucus, which prevents them from drying out. Sometimes snails are also able to hibernate in the summer to survive if they are faced with a severe drought.

- **Forms**
- **Mechanism**
- **Techniques**
- **Relationships**
- Each of these embodies fundamental design strategies**.** These are known as ***Design Paradigm*- such**
- **as, *nesting, smaller and larger, bigger and smaller, taller and shorter,* etc. are some of the design**
- **paradigm.**

- In nature each form has a role to play. Each form has functional mechanism. The mechanism has built in technique which relates to the entire body of the living being. Could you think of any part of human being does not have role to play to the entire body? However, over the period of time the functioning of some of the organs might have become irrelevant or weaker because of the changing living conditions.

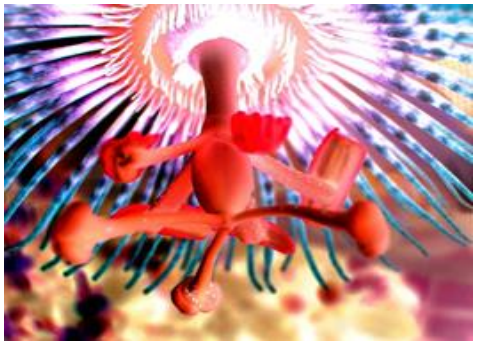
Space Capsule



Find out the relevance of the followings-
Forms
Mechanism
Techniques
Relationships

Flora and Fauna of Colombia

Colombia has the largest number of species that are not found naturally anywhere in the worldwide. About 10% of the species in the world live in Colombia

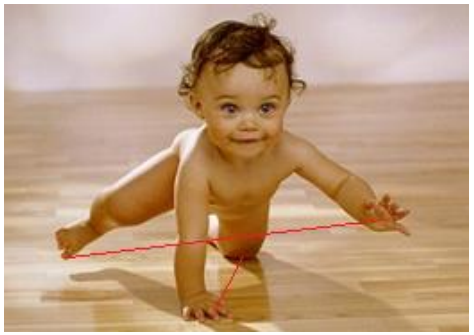


Animal Colors

There are several separate reasons why animals have evolved colours-

- **Camouflage** enables an animal to remain hidden from view;
- **Signalling** enables an animal to communicate information such as warning of its ability to defend itself;
- **Color in advertising**, animals also use signalling services to animals of other species;
- To **signal sexual status** to other members of the same species; and
- **Color to divert attacks,** some animals use by sudden movement, surprising a predator e.g. with eyespots or other flashes of colour, and possibly by motion dazzle, **confusing a predator's attack by moving a bold pattern** (such as zebra stripes) rapidly. Some animals are **coloured for physical protection**, such as having pigments in the skin to protect against sunburn, while some frogs can lighten or darken their skin for temperature regulation.
- Finally, **<u>animals can be coloured incidentally</u>**. For example, blood is red because the haem pigment (red organic pigment) needed to carry oxygen is red. Animals coloured in these ways can have striking natural patterns.

Why do we want to mimic- trying to emulate animals

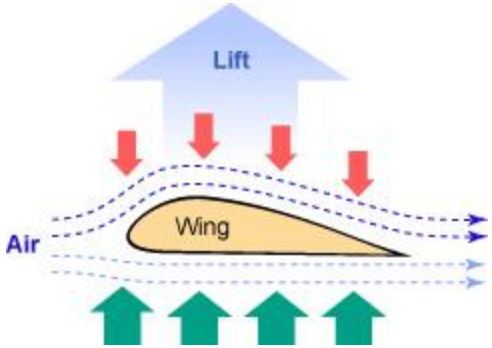


Scientists trying to figure out how cheetahs manage to run faster than any other mammal might have a few tips to pass along to human runners, reports the *New York Times*'. For one thing, build up those thighs. The cheetah has powerful upper legs, which allow them to pump faster than, say, a greyhounds' legs. (The two animals were compared in the experiment because they're similar in build, but wild cheetahs top out at 65mph and the dogs at 40mph. The fastest humans clock in around 28mph.) (Ref. http://www.newser.com/story/148995/to-run-like-a-cheetah-get-stronger-thighs.html Feb 25th 2016)

What is ‘Design Paradigm’?

Design Paradigm presents-

- **A new approach to solving design problems and thinking about the ways things work**
- **How things work?**
- **It is a powerful tool for ‘innovation and creative work’.**



How avian flies? What makes their flight comfortable and easy to fly?

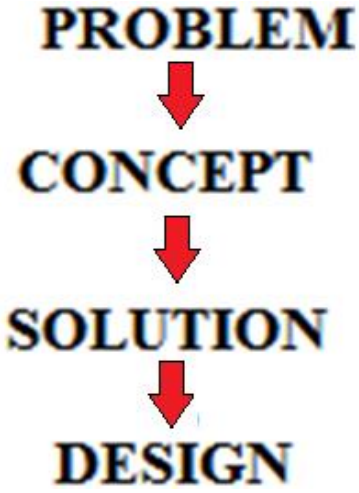
Capturing the air in the wings to fly/ sail; the direction of the wind helps to fly comfortably.
https://youtu.be/hmN11jZd3bU (Bird Fly)

Plate 1A Aves

1B Flying Suit (manmade)

1C Flying Squirrel

- The flying of *avis or Aves* (plate 1A) by capturing air in the wings and other similar species (plate 1C) has inspired man to invent flying devices (plate 1B) to fulfill their aspirations in innovative manner. The observation of flying creatures in nature has inspired man to think differently. The inquisitive mind wanted to know the ways things work and came up with possible solutions.
- Therefore, Design Paradigm searches for solutions by observing nature. It spans to help wide range of areas for the problem solving. Designer has to first understand the nature of 'problem' before searching for the solution. While trying to understand 'paradigm' the course would try to focus on its relationship to design and application for innovations. Therefore, it is important to address the following issues before we proceed further-



Faced a problem

- Define/ analyze
- Concept is generated
- Solution is found
- Design (Prototype)
- Testing

It was invented 30 years ago on February 10, 1983 when **Carmela Vitale** got her patent issued for that piece of plastic. That tiny little plastic invention has saved humanity from eating smashed and stripped pizza that has half of its cheese ripped off because the pizza got stuck to the cardboard top of the pizza box.

Pizza Saver

- In most cases we have the tendency to copy from another product. Without confronting a ‘problem’ your design thinking is clouded by the influence of another product. Therefore, you need to search for the problem that you have observed or confronted. Thereafter the concept is generated and search for a solution is found.

VID-20170221-WA0023.mp4

- Let us think about an exercise-
- My house remains static facing the directions same years after year
- I would like to see my bedroom, drawing-room, kitchen, study-room, etc. face all the directions
- In modern days is it possible to choose and buy an apartment or build a house that would fulfil the above desire?
- Urban landscape has limited land space
- It is a problem (identified)
- Where is the solution?
- How do I design to address the above problems?
- Problem is identified, now you need to think about a possible solution
- The ‘concept’ has to be generated to solve the problem
- The most important step is to design the ‘concept’ with logical solution
- It could be an ‘out of the box’ concept

Duck family crossing the road

BLOCK BLOCK

How to create an impression on a 2D surface?

Climbing a tall coconut tree

Try to address the problem given bellow

How do you know or understand Problem?

Designers should ask the following fundamental questions-

- What is the essence of the ‘problem’?
- Where is the essence of the prior solution (if any)?
- Where did the prior solution fail?
- If it did not fail, how can the prior solution is improved upon?
- What other ways might there be to do things?
- Are there other problems similar to this one?
- What form does this solution imply?
- What other forms or basic approaches might work as well?

Geodesic Dom Structure

- **How is Design Paradigm related to design education?**

- Every budding design student needs to remember that Design Education cannot be confined inside the studio. The nourishment for innovative thinking and searching for new ideas can only be possible provided they are exposed to the splendid of nature's creation. Every well-known designer (Buckminster Fuller, Frank Lloyd Wright, etc.) have paid special attention toward nature. The Paradigm in nature has attracted human beings for thousands of years. It is well accepted that nature is the greatest teacher in the field of design. Designers without having enough exposure and observation of nature would unfortunately lead to excellent craftsmen but would fail to make innovative solutions. Therefore, it is absolutely necessary for the design students to get closer to nature's wealth. The solutions of nature's problem would encourage towards innovative ideas. Buckminster Fuller's invention of Geodesic Dom is based on his observation of Beehive Structure. In the process the same structure has led to invention of Modular Structures, which has resulted in revolutionized the modern day design concepts. Today no design educator would deny the role of nature's paradigm in the design education.

Have you observed behavior of living beings around you? Have you observed their movements? How many types of movement have you observed?

Crawlling Insect

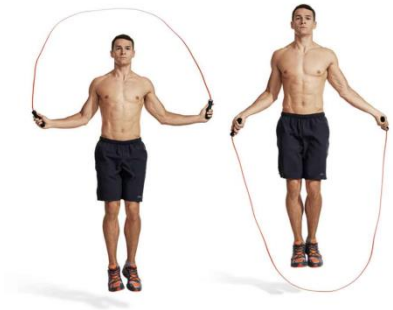
Flying Insect

Jumping Animal. Do they jump in the same way?

- **Does Design Paradigm help us to think differently?**
- Nature's Paradigms are the benchmark of inspirations that mankind has tried to used as tool for innovations. It teaches us to think differently. Design Paradigm or the study of Biomimicry is a powerful tool for innovative ideas. It helps to question the fundamentals of functioning. **It is to discovering of a fundamental truth about the way things work.** <u>The paradigm has given us a new way to categorize things we encounter and find a new approach to formulating new ideas</u>.

There is one common ability to all the above, without that none can function.

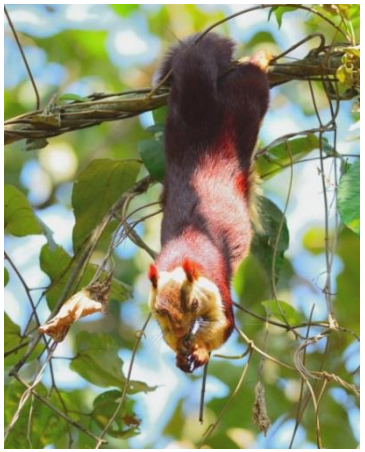
How many ways human being can jump? Each time it shows new gesture for a specific purpose.



During the jump every time observe the coordination between the limbs. Through the coordination we try to maintain our balance naturally.

Balancing Human being

Balancing



Try to observe how each one is trying to maintain the balance (equilibrium). Do you know what maintains your balance?

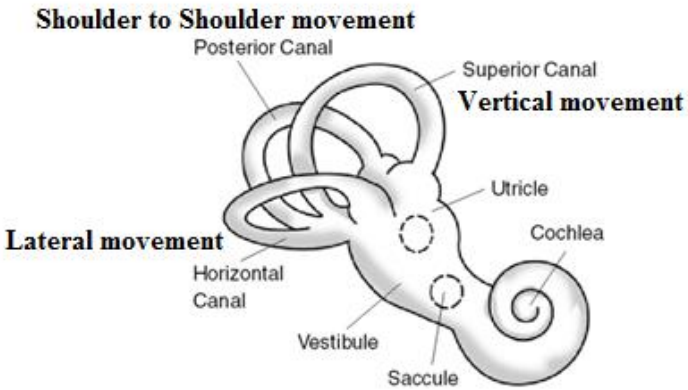
Type of Balance:

1. **Visual Balance**
2. **Physical Balance**

- **What is balance (Physical Balance)?**
- Balance is the ability to maintain the body's center of mass over its base of support with help of various sensory organs.
- A properly functioning balance system allows humans to perform the followings-

- **Clarity in movement**
- **Identify orientation with respect to the physical positioning**
- **Determine direction and speed movement**
- **Make automatic adjustment to maintain posture and stability**

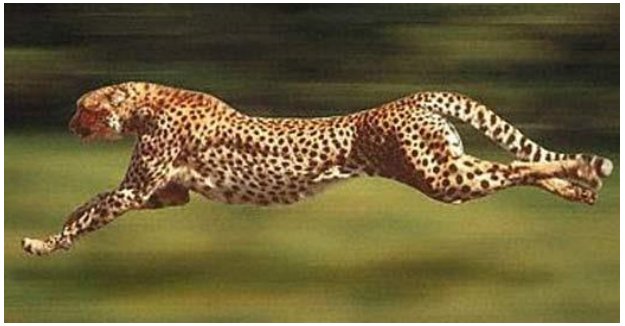
- (Source: https://vestibular.org/understanding-vestibular-disorder/human-balance-system (Feb. 27, 2017)



Paradigms are all around us. Why don't you enjoy discovering it? If something is known (not in real sense) to us we do not want to discover further consciously. We do not want to know the science behind the phenomenon until it is taught in the classroom. Why don't you question? It shouldn't be always part of the classroom learning. Because the classroom learning is always considered boring. Isn't it?

Source: https://www.reference.com/science/part-body-controls-balance-193b8c53f2796a3# (Feb. 27, 2017)

Speed- Running



Flying

Climbing

Running Cheetah & Horse; flying bat & squirrel; climbing monkey & bear and goat on a tree